1.Database

1.1 Suggestions on Storage of JT808 Standard Data Collection

The data collection at the terminal level will increase linearly with the number of terminals. Conventional relational databases are not very capable of meeting the requirements for time-series data storage and retrieval performance, especially in terms of scalability

It is recommended to use ClickHouse for the storage of JT808 data collection. It offers excellent read and write performance, a high data compression ratio to save storage costs, and is easy to scale horizontally.

1.2 Clickhouse JT808 Database Table Design

GPS Point Table：

```
CREATE TABLE vehicle_gps ON CLUSTER 'ck_1_shard_2_replicas'
(

    `vehicle_id` String COMMENT 'Vehicle ID,
   `device_id` String COMMENT 'device_id',
    `group_id` String COMMENT 'group_id',
   `protocol_id` String COMMENT 'protocol_id,Private/JT808 Inconsistency',
   `protocol_type` UInt8 COMMENT 'protocol_type',
   `vin` String COMMENT 'VIN" (Vehicle Identification Number)',
   `manufactor` Nullable(UInt32) COMMENT 'Manufacturer Code',
   `software_version` String COMMENT 'software_version',
   `mcu_version` String COMMENT 'mcu_version',
   `hardware_version` String COMMENT 'hardware_version',
   `algorithm_version` String COMMENT 'algorithm_version',
   `sn` String COMMENT 'terminal sn',
```

```
    `device_type` UInt8 COMMENT 'device_type',
    `sim` String COMMENT 'sim sn',
    `driver_id` String COMMENT 'driver_id',

    `gps_time` DateTime64(3, 'Asia/Shanghai'),
    `longitude` Decimal64(8) COMMENT ' Longitude, with 8 decimal places preserved.
    `latitude` Decimal64(8) COMMENT ' Latitude, with 8 decimal places preserved.
    `accuracy` Nullable(Float32) COMMENT ' Location accuracy, in meters
    `altitude` Nullable(Float32) COMMENT ' altitude,meters
    `gps_speed` Float32 COMMENT 'gps_speed,km/h',
    `direction` Nullable(Float32) COMMENT 'GPS heading angle',
    `state_bit` Nullable(Int32) COMMENT 'Vehicle Status Bit',
    `warn_bit` Nullable(Int32) COMMENT 'warning bit',
    `location_state` Nullable(UInt8) COMMENTGPS positioning status, 0. Unlocated 1. Located',
    `acc` Nullable(Int8) COMMENT 'acc status，0.close 1.open',
    `mileage` Nullable(Float64) COMMENT 'Total Mileage, Unit Meters',
    `oil` Nullable(Float64) COMMENT 'Fuel Level, Unit Liters',
    `recorder_speed` Nullable(Float32) COMMENT 'Speed obtained from driving record
function',
    `temperature` Nullable(Int32) COMMENT 'Carriage Temperature',
    `gnss_num` Nullable(UInt8) COMMENT 'GNSS Number of Satellites',
    `network_intensity` Nullable(Int32) COMMENT 'Wireless Network Strength, Unit dBm value',
    `tire_pressure` Array(Nullable(UInt8)) COMMENT 'Tire Pressure Value List, Unit Pa (Pascal)',
    `signal_state_bit` Nullable(Int32) COMMENT 'Signal Status Bit, see Table 4 for details',
    `io_state_bit` Nullable(Int32) COMMENT 'IO Status Bit, see Table 5 for details',
    `analog_quantity` Nullable(Int32) COMMENT 'analog_quantity',
    `load_weight` Nullable(UInt8) COMMENT 'load_weight',
    `video_warn_bit` Nullable(Int32) COMMENT 'Video warning Bit',
    `video_lost_warn_bit` Nullable(Int32) COMMENT 'Video Signal Loss warning Bit',
    `video_occ_warn_bit` Nullable(Int32) COMMENT 'Video Signal Blockage warning Bit',
    `storage_warn_bit` Nullable(Int32) COMMENT 'Storage Malfunction warning Bit',
    `strategy_bit` Nullable(Int32) COMMENT '',
    `resume` UInt8 DEFAULT 0 COMMENT 'Re-transmission Flag, 0. Real-time 1.
Re-transmission',

    `city_code` String COMMENT 'Administrative Division - City',
    `city` String COMMENT 'city',
    `province_code` String COMMENT 'Administrative Division - Province',
    `province` String COMMENT 'Province Name',
    `weather` String COMMENT 'Weather Description (Sunny, Partly Cloudy, Mostly Sunny,
Overcast, etc.',
    `winddirection` String COMMENT '"Wind Direction',
    `windpower` String COMMENT 'Wind Speed',

    `weather_temperature` Nullable(Int8) COMMENT 'Temperature',
```

```
    `humidity` Nullable(Int8) COMMENT 'Humidity',
    `weather_reporttime` DateTime64(3, 'Asia/Shanghai') COMMENT 'Report Time',

    `ip_address` String COMMENT 'Source IP Address, ip:port',
    `receive_time` DateTime64(3, 'Asia/Shanghai') COMMENT 'Message receive time',
    `create_at` DateTime DEFAULT now() COMMENT 'create time'
  )
ENGINE = ReplicatedReplacingMergeTree()
PARTITION BY toYYYYMM(gps_time)
PRIMARY KEY (vehicle_id, gps_time)
ORDER BY (vehicle_id, gps_time)
TTL toDate(gps_time) + INTERVAL 365 DAY DELETE
SETTINGS index_granularity = 8192
COMMENT 'Vehicle GPS Table';
```

Terminal Alarm Table:

```
CREATE TABLE vehicle_alarm ON CLUSTER 'ck_1_shard_2_replicas'
(
    `vehicle_id` String COMMENT 'Vehicle ID',
    `device_id` String COMMENT 'Device ID',
    `device_type` UInt8 COMMENT 'device type',
    `group_id` String COMMENT 'group id',
    `protocol_id` String COMMENT 'protocol_id,Private/JT808 Inconsistency',
    `protocol_type` UInt8 COMMENT 'Protocol Type, see Protocol Type Table 1',
    `vin` String COMMENT 'VIN" (Vehicle Identification Number)',
    `manufactor` Nullable(UInt32) COMMENT 'Manufacturer Code',
    `software_version` String COMMENT 'software version',
    `mcu_version` String COMMENT 'mcu version',
    `hardware_version` String COMMENT 'hardware version',
    `algorithm_version` String COMMENT 'algorithm version',
    `sn` String COMMENT 'device sn',
    `sim` String COMMENT 'sim sn',
    `driver_id` String COMMENT 'driver ID',
```

```
`gps_time` DateTime64(3, 'Asia/Shanghai'),
`longitude` Decimal64(8) COMMENT '"Longitude, with 6 decimal places preserved.',
`latitude` Decimal64(8) COMMENT '"Latitude, with 6 decimal places preserved.',
`accuracy` Nullable(Float32) COMMENT 'Positioning accuracy, meters',
`altitude` Nullable(Float32) COMMENT 'altitude,meters',
`gps_speed` Float32 COMMENT 'GPS vehicle speed, km/h',
`direction` Nullable(Float32) COMMENT 'GPS heading angle',
`state_bit` Nullable(Int32) COMMENT 'Vehicle status bit, see Table 2 for details on status bits',
`warn_bit` Nullable(Int32) COMMENT 'warning identification, see Table 3 for details on warning flags',
`mileage` Nullable(Float64) COMMENT 'Accumulated mileage, unit meters',
`oil` Nullable(Float64) COMMENT 'Fuel Level, Unit Liters',
`recorder_speed` Nullable(Float64) COMMENT 'Speed obtained from driving record function.',
`temperature` Nullable(Int32) COMMENT 'Carriage temperature',
`gnss_num` Nullable(UInt8) COMMENT 'Number of GNSS satellites',
`network_intensity` Nullable(Int32) COMMENT 'Wireless network strength, unit dBm value',
`tire_pressure` Array(Nullable(UInt8)) COMMENT 'Tire pressure values list, unit Pa',
`signal_state_bit` Nullable(Int32) COMMENT 'Signal status bit, see Table 4 for details',
`io_state_bit` Nullable(Int32) COMMENT 'I/O status bit, see Table 5 for details',
`analog_quantity` Nullable(Int32) COMMENT 'analog quantity',
`load_weight` Nullable(UInt8) COMMENT 'load weight',
`video_warn_bit` Nullable(Int32) COMMENT 'video warning bit',
`video_lost_warn_bit` Nullable(Int32) COMMENT 'video lost warning bit',
`video_occ_warn_bit` Nullable(Int32) COMMENT 'Video Signal Blockage warning Bit',
`storage_warn_bit` Nullable(Int32) COMMENT 'Storage Malfunction warning Bit',
`strategy_bit` Nullable(Int32) COMMENT '',

`alarm_id` String COMMENT 'Platform alarm ID (global)',
`jt_alarm_id` String COMMENT 'jt808 alarm ID',
`alarm_start_time` DateTime64(3, 'Asia/Shanghai'),
`alarm_end_time` DateTime64(3, 'Asia/Shanghai'),
`alarm_type` Int32 COMMENT 'alarm type',
`alarm_level` UInt16 COMMENT 'alarm level',
`alarm_sign` UInt16 COMMENT 'alarm sigh',
`front_speed` Nullable(Int32) COMMENT 'Speed of the preceding vehicle',
`front_distance` Nullable(Int32) COMMENT 'Distance to the preceding vehicle/pedestrian',
`ldw_type` Nullable(UInt8) COMMENT '',
`road_sign` Nullable(UInt32) COMMENT '',
`road_sign_value` Nullable(UInt32) COMMENT '',
`fatigue_degree` Nullable(UInt32) COMMENT '',
```

```
    `area_type` Nullable(UInt16) COMMENT '',
    `area_id` Nullable(UInt32) COMMENT '',
    `area_direction` Nullable(UInt8) COMMENT '',
    `area_drive_time` Nullable(UInt32) COMMENT '',
    `area_drive_result` Nullable(UInt32) COMMENT '',
    `file_total` UInt32 COMMENT 'total attachments',

    `ip_address` String COMMENT 'Source IP address, IP:port',
    `receive_time` DateTime64(3, 'Asia/Shanghai') COMMENT 'Server receive time',
    `resume` UInt8 DEFAULT 0 COMMENT 'Re-transmission flag, 0. Real-time, 1. Re-transmission',
    `visible` UInt8 DEFAULT 1 COMMENT 'Visual status, 0. Not visible, 1. Visible',
    `deal` UInt8 DEFAULT 0 COMMENT 'Processing status, 0. Unprocessed, 1. Processed',
    `false_detection` UInt8 DEFAULT 0 COMMENT 'False alarm identification, 0. Unprocessed, 1. True alarm, 2. False alarm',
    `info_process` Int8 DEFAULT 0 COMMENT 'Info secondary processing flag, 0. Unprocessed, 1. Processed, -1 File timeout, -2 Processing timeout',
    `tags` Array(String) COMMENT 'Tag array.',
    `create_at` DateTime DEFAULT now() COMMENT 'create time'
  )
ENGINE = ReplicatedReplacingMergeTree()
PARTITION BY toYYYYMM(alarm_start_time)
ORDER BY (vehicle_id, alarm_start_time, alarm_type, alarm_id)
TTL toDate(alarm_start_time) + INTERVAL 365 DAY DELETE
SETTINGS index_granularity = 8192
COMMENT 'Vehicle alarm table.';
```

Terminal alarm file index table:

```
CREATE TABLE vehicle_alarm_file ON CLUSTER 'ck_1_shard_2_replicas'
(
    `vehicle_id` String COMMENT 'Vehicle ID',
    `device_id` String COMMENT 'Device ID',
```

```sql
	`group_id` String COMMENT 'Group id',
	`device_type` UInt8 COMMENT 'device type',
	`protocol_id` String COMMENT 'Protocol ID',
	`protocol_type` UInt8 COMMENT 'Protocol types: 1. MQTT 2. JT808 standard',
	`vin` String COMMENT 'VIN (Vehicle Identification Number)',
	`manufactor` Nullable(UInt32) COMMENT 'Manufacturer Code',
	`software_version` String COMMENT 'software_version',
	`mcu_version` String COMMENT 'mcu_version',
	`hardware_version` String COMMENT 'hardware_version',
	`algorithm_version` String COMMENT 'algorithm_version'',
	`sn` String COMMENT 'terminal sn',
	`sim` String COMMENT 'sim sn',
	`driver_id` String COMMENT 'Driver ID',

	`alarm_type` UInt16 COMMENT 'alarm type',
	`alarm_start_time` DateTime64(3, 'Asia/Shanghai'),
	`alarm_id` String COMMENT 'Alarm unique identifier',
	`jt_alarm_id` String COMMENT 'JT808 alarm sign',
	`channel` UInt8 COMMENT 'channel',
	`format` UInt8 COMMENT 'format',
	`index_num` UInt8 COMMENT 'sequence',
	`file_name` String COMMENT 'file name',
	`size` UInt64 COMMENT 'file size',
	`url` String COMMENT 'file url',
	`in_url` String COMMENT 'intranet url',
	`path` String COMMENT 'Relative path',
	`snapshot_url` String COMMENT 'Preview URL',
	`play_url` String COMMENT 'Video playback URL',
	`upload_time` DateTime64(3, 'Asia/Shanghai') COMMENT 'File upload time',

	`ip_address` String COMMENT 'Source IP address, IP:port',
	`receive_time` DateTime64(3, 'Asia/Shanghai') COMMENT 'Server receive time',
	`visible` UInt8 DEFAULT 1 COMMENT 'Visibility status, 0. Not visible 1. Visible',
	`tags` Array(String) COMMENT 'Tags Array',
	`source_by` UInt8 DEFAULT 1 COMMENT 'Source, 1. Device upload 2. Platform secondary processing generated',
	`create_at` DateTime DEFAULT now() COMMENT 'create time'
    )
ENGINE = ReplicatedReplacingMergeTree()
PARTITION BY toYYYYMM(alarm_start_time)
ORDER BY (vehicle_id, alarm_start_time, alarm_type, format, index_num)
TTL toDate(alarm_start_time) + INTERVAL 365 DAY DELETE
SETTINGS index_granularity = 8192
COMMENT 'Local table of vehicle alarm attachments';
```